STAFAC 500 g/kg Powder

FORMULA:

Virginiamycin : 50% Activity
Each 1 kg of premix contains 500 g Virginiamycin activity

PRODUCT DESCRIPTION:

It is a free flowing fine granulated beige powder.

PHARMACODYNAMICS:

The target of both components of virginiamycin is the 50S subunit of the bacterial ribosome.

The A-component (virginiamycin M1) binds to a tight pocket within the peptidyl transferase catalytic centre (PTC) of the ribosome whereby the attachment of transfer RNA (tRNA) to both the acceptor (A) site and the donor (P) site of the PTC is prevented. This leads to a shutoff of peptide bond formation and stops the elongation of the growing polypeptide chain. This binding site and inhibition mechanism is similar to that of certain other antibiotics, such as chloramphenicol for example. In contrast, the B component (virginiamycin S1) does not affect the peptidyl transferase reaction but inhibits the peptide elongation process after a few cycles of peptide bond formation by binding to the 23S ribosomal RNA within the ribosomal exit tunnel. This is similar to the mode of action of macrolide antibiotics, such as erythromycin. Hereby, extension of the nascent protein chain is prevented and additionally the peptidyl-tRNA is released from the ribosome, which results in incomplete peptides.

The structural composition of the ribosomal complex is important as virginiamycin S1 acts synergistically to the conformational change of the peptidyl transferase centre of the 50S-ribosome induced by virginiamycin M1.

Individually, each component of virginiamycin exhibits only moderate bacteriostatic activity by binding to the 50S subunit of bacterial ribosomes, thereby blocking translational processes, whereas the combination of the two components leads to a synergistic effect with a hundred fold higher activity that results in a bactericidal activity.

As the mechanism of action of both factors is different, there are several possible mechanisms of resistance.

PHARMACOKINETICS:

Studies were performed to extract and quantify total radioactive residues in liver and urine from rats administered 14C-virginiamycin by gavage (0 and 25 mg/kg bw/day for 14 days; 7 animals/sex/group) and to fractionate the extracts using high performance liquid chromatography (HPLC) and thin layer chromatography (TLC) in order to determine the extent of virginiamycin fragmentation. The majority of liver residues were considered non-extractable. Fractionation of tissue residues by HPLC indicated that no single component represented more than 3.3% of total liver residues. In urine, no single fraction (HPLC) represented more than 9.9%, and no region (TLC) represented more than 5.7% of the total urine residues.

The metabolites in liver from rat, turkey and cattle treated orally with 14C-virginiamycin were compared. Results demonstrate a similar metabolic profile in turkeys, cattle and rats, indicating that the rat is a suitable model for comparative purposes. The majority (43-65%) of the total liver residue from virginiamycin treated turkeys, cattle, and rats was found to be non-extractable (present as a bound residue). The extractable residues were shown to consist of multiple components indicating that virginiamycin is extensively metabolised in the tissue.

INDICATION:

Chickens (Broilers)- Prevention of necrotic enteritis caused by Clostridium perfringens susceptible to virginiamycin in broiler chickens.
Swine - Treatment of swine dysentery in non-breeding swine weighing over 120 lb (54.43 kg)
Swine - Treatment and control of swine dysentery in swine weighing up to 120 lb (54.43 kg)
Swine - Aid in control of swine dysentery in swine weighing up to 120 lb (54.43 kg).
For use in animals or on premises with a history of swine dysentery but where symptoms have not yet occurred.

RECOMMENDED DOSAGE:

Chickens (Broilers)	Required Level of Virginiamycin per 1000 kg of Complete Feed	Amount of Stafac 500 per 1000 kg of Complete Feed
Prevention of necrotic enteritis caused by Clostridium perfringens susceptible to virginiamycin	22 g	44 g

Swine	Required Level of Virginiamycin per 1000 kg of Complete Feed	Amount of Stafac 500 per 1000 kg of Complete Feed
Treatment of swine dysentery in nonbreeding swine weighing over 120 lb (54.43 kg)	110g for 2 weeks	220g
Treatment and control of swine dysentery in swine weighing up to 120 lb (54.43 kg)	110 g for 2 weeks follow by 55 g for 4 weeks	220 g 110 g
Aid in control of swine dysentery in swine weighing up to 120 lb (54.43 kg). For use in animals or on premises with a history of swine dysentery but where symptoms have not yet occurred	27.5 g	55 g

MODE OF ADMINISTRATION:

Oral administration via feed.
Mixing Directions: Determine the appropriate amount of Stafac 500 required from the table under recommended dosage. Mix thoroughly in feed.

CONTRAINDICATIONS:

Not for use in laying chickens.

WARNINGS AND PRECAUTIONS:

(1) Do not feed to replacement or breeding chickens.
(2) Do not use in feeds containing pellet-binding agents with the exception of bentonite, Pel-Aid, Agri-Colloid, and Lignosol.
(3) Do not feed undiluted.
(4) Not for use in breeding swine over 120 pounds.

INTERACTIONS WITH OTHER MEDICAMENTS:

Not known.

STATEMENT ON USAGE DURING PREGNANCY AND LACTATION:

Not for use in laying chickens.

ADVERSE EFFECTS / UNDESIRABLE EFFECTS:

Not known.

OVERDOSE AND TREATMENT:
Not known. If overdose is confirmed, remove the feed containing virginiamycin and provide proper access to drinking water to the birds/swine.

WITHDRAWAL PERIOD(S):
The withdrawal period is 0 (zero) days

STORAGE CONDITIONS:
Store in a cool dry place below 25°C. Keep bag closed when not in use.

SHELF LIFE:
3 years.

SHELF LIFE AFTER RECONSTITUTION OF PRODUCT INTO FEED:
2 months.

DOSAGE FORMS AND PACKAGING AVAILABLE:
Medicated powder premix
25 kg bag

Manufactured by Bio Agri Mix LP
52 Wellington Street Mitchell, ON N0K 1N0, Canada

Registration holder: Phibro Corporation (M) Sdn Bhd
311, Level 3, Menara Axis,
2, Jalan 51A/223, Seksyen 51A,
46100 Petaling Jaya, Selangor, Malaysia

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